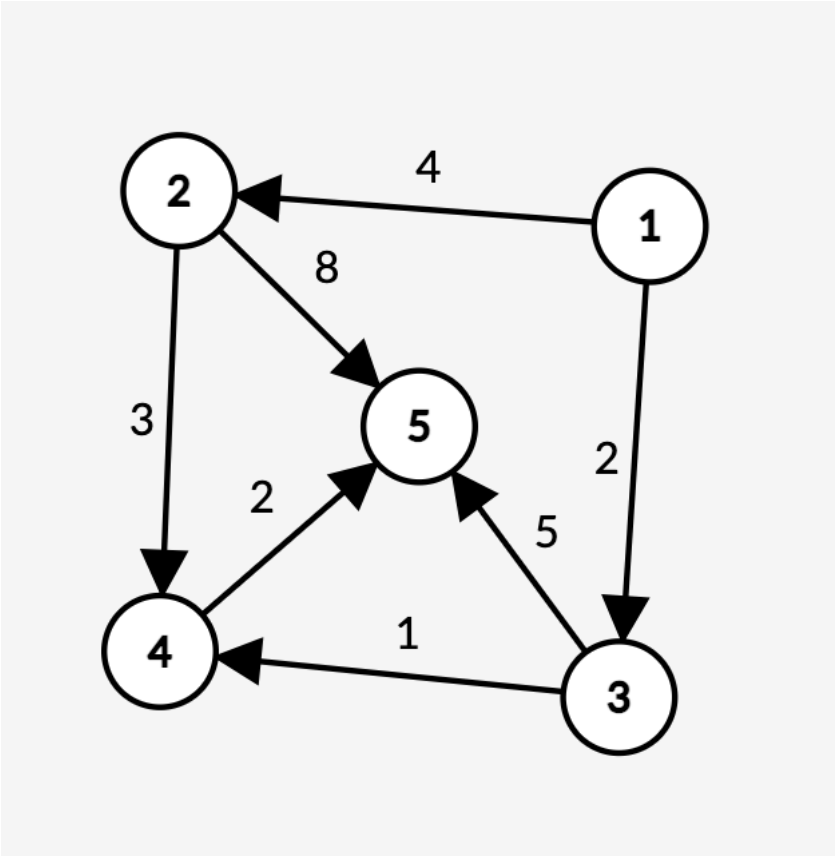


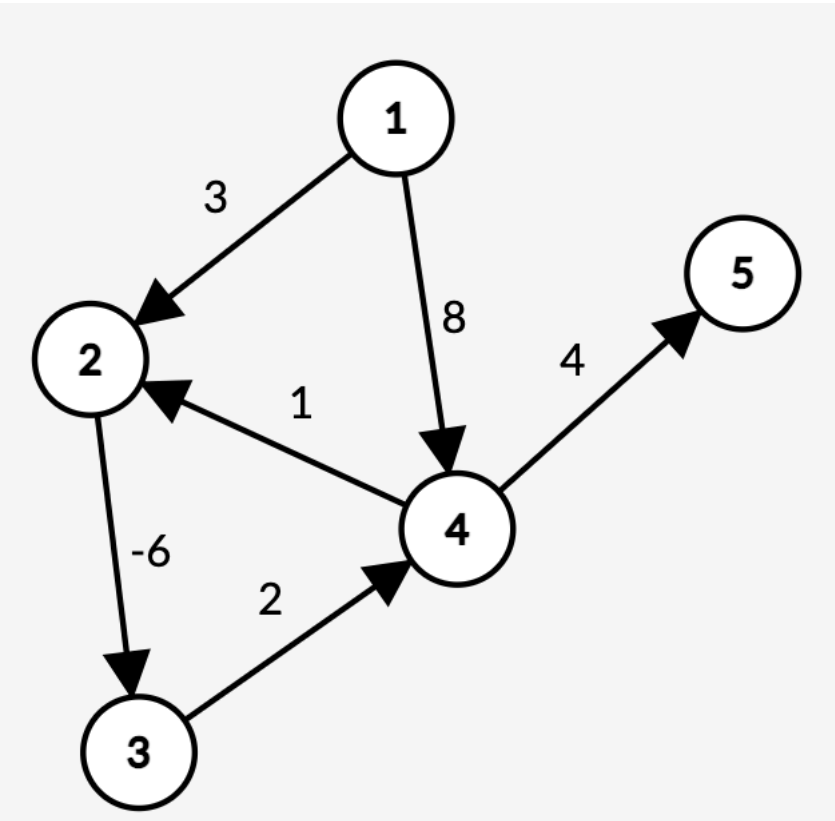
Bellman ford - exercitii Claude

1. nod de start = 1

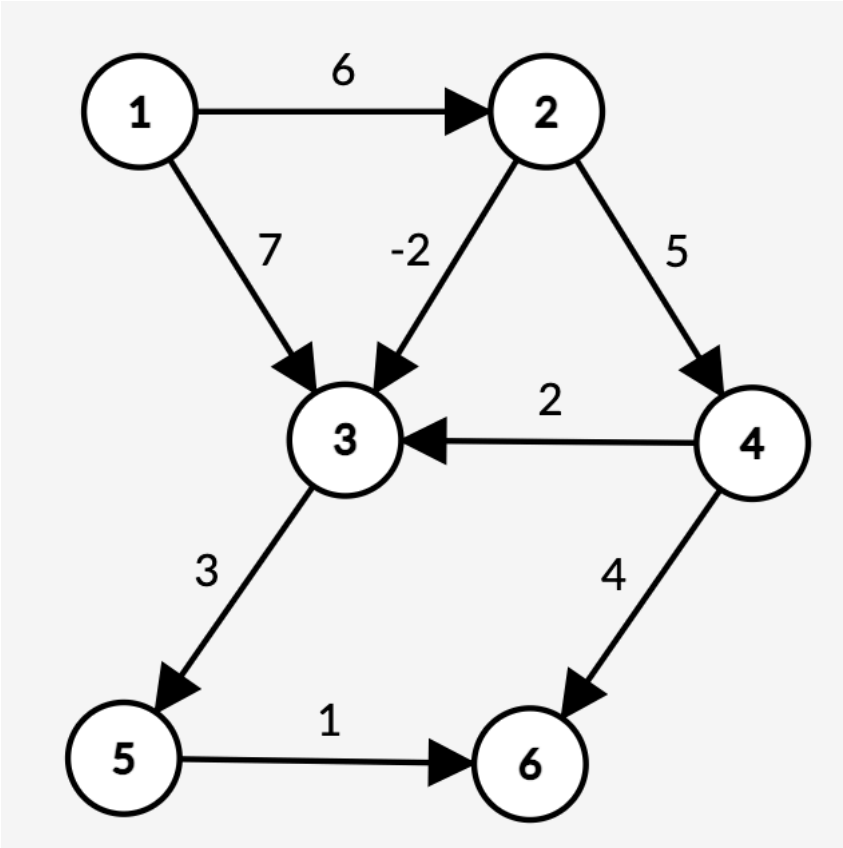


2. nod de start

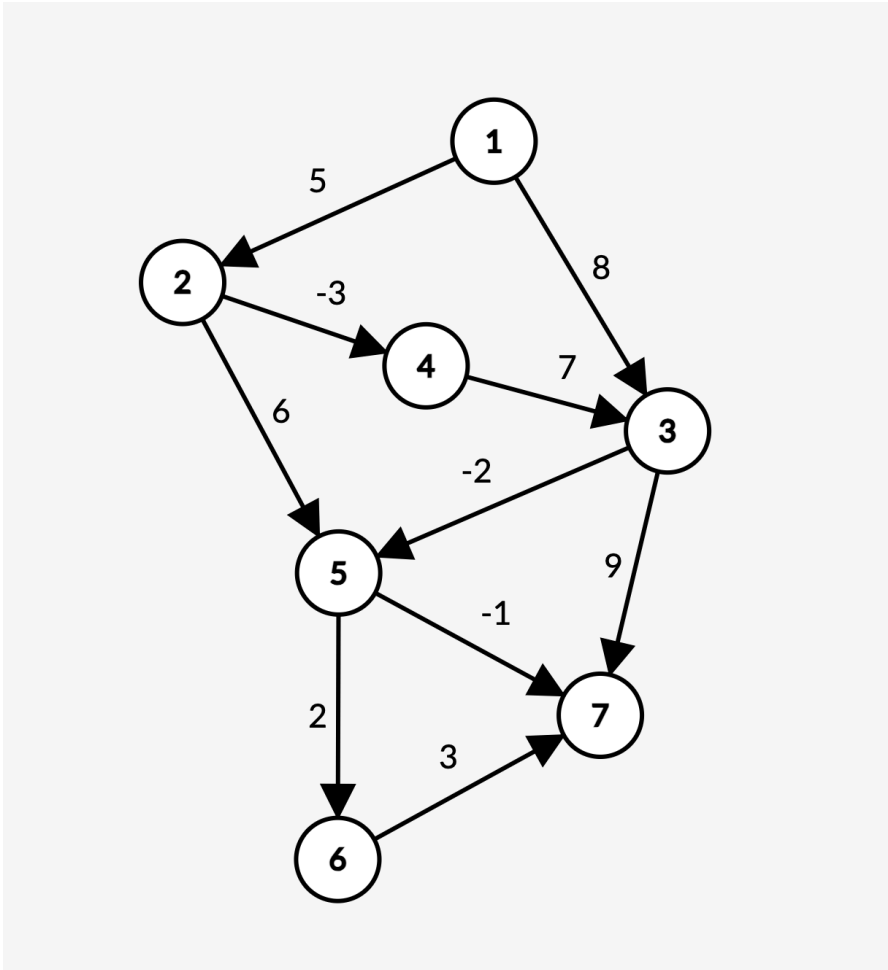
= 1



3. nod de start = 1

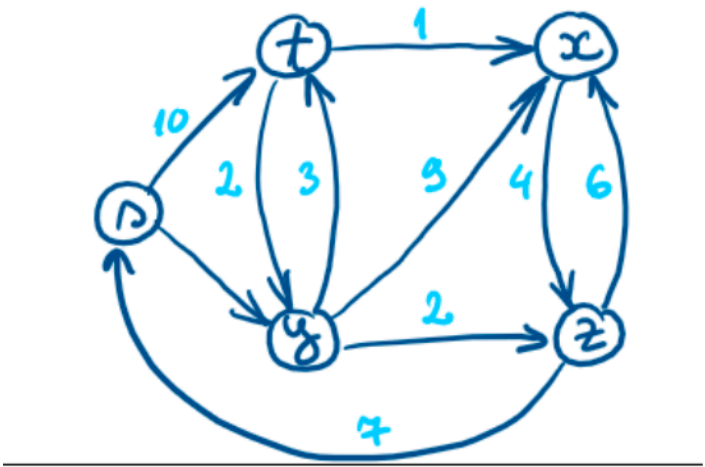


4.
nod de start 1



din seminar 2DIJKSTRA

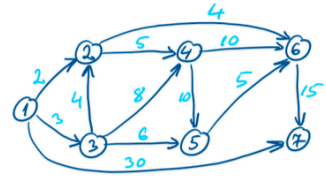
Tabela 2: Valoarea atributelor d și π la finalul execuției algoritmului pentru graful dat la problema 3.
s t x y z
0 8 9 5 7
NIL y t s y



$$V_i^{(k)} = \min_{j=1, n} \{a_{ij} + V_j^{(k-1)}\}, \text{ pentru } i = 1, 2, \dots, n$$

până când $V^{(t)} = V^{(t-1)}$ pentru un t .

Problema 4. Pentru graful alăturat să se determine folosind algoritmul lui Bellman-Kalaba distul minim pentru drumurile care duc în vârful 7.



Soluție Matricea de adiacență asociată grafului este

$$A = \begin{pmatrix} 0 & 2 & 3 & \infty & \infty & \infty & 30 \\ \infty & 0 & \infty & 5 & \infty & 4 & \infty \\ \infty & 4 & 0 & 8 & 6 & \infty & \infty \\ \infty & \infty & \infty & 0 & 10 & 10 & \infty \\ \infty & \infty & \infty & \infty & 0 & 5 & \infty \\ \infty & \infty & \infty & \infty & \infty & 0 & 15 \\ \infty & \infty & \infty & \infty & \infty & \infty & 0 \end{pmatrix}$$

iar vectorii distanță sunt

$$V^{(1)} = \begin{pmatrix} 30 \\ \infty \\ \infty \\ \infty \\ \infty \\ 15 \\ 0 \end{pmatrix}, V^{(2)} = \begin{pmatrix} 30 \\ 19 \\ \infty \\ 25 \\ 20 \\ 15 \\ 0 \end{pmatrix}, V^{(3)} = \begin{pmatrix} 21 \\ 19 \\ 23 \\ 25 \\ 20 \\ 15 \\ 0 \end{pmatrix}, V^{(4)} = \begin{pmatrix} 21 \\ 19 \\ 23 \\ 25 \\ 20 \\ 15 \\ 0 \end{pmatrix}.$$